

Data Science

Data Science is a combination of multiple disciplines that uses statistics, data analysis, and machine learning to analyze data and to extract knowledge.

Data Science is about data gathering, analysis, and decision-making.

By using Data Science, companies are able to make:

- * Better decisions.
- * Predictive analysis.

Data Science Needed:

* Data Science is used in many industries in the world today, e.g.: banking, consultancy, healthcare and manufacturing.

* For route planning: To discover the best routes to ship.

* To foresee delays for flight/ship/train etc. (through predictive analysis).

* To create promotional offers.

* To find the best suited time to deliver goods.

* To forecast the next years revenue for a company.

* To analyze health benefit of training.

* To predict who will win elections.

Data science can be applied in

Consumer goods

Stock Markets

Industry

Politics

Logistic Companies

E-commerce.

Data Scientist Work?

* Machine Learning.

* Statistics

* Programming

* Mathematics

* Databases.

1) They find patterns within the data. And they organize the data in a standard format.

Data Scientist Works:

- 1) Ask the right questions.
- 2) Explore and collect data.
- 3) Extract the data.
- 4) Clean the data.

Errors on Missing values

Find and replace missing values.

Normalize data.

Analyze the data

- 1) Represent the result.

Data Science:

It is a field that blends scientific methods; applies Business analysis, Math, Programming and statistics skills

Collected and Analyzed data → Data Translator → Relevant Information and useful insights.

Properties of Data - Big Data.

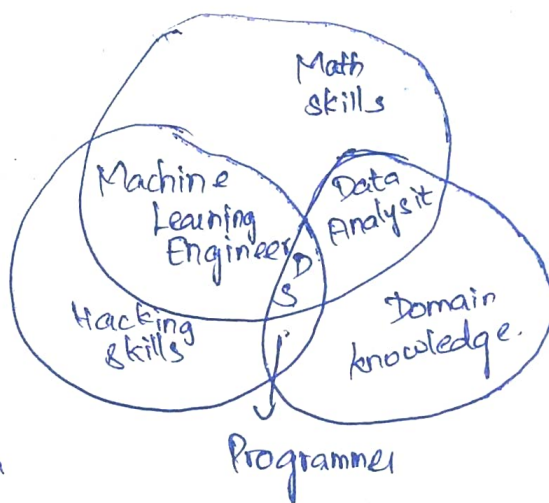
There are five v's of Big data that explains the characteristics.

* 5 V's of Big Data

- 1) Volume
- 2) Variety.
- 3) Veracity.
- 4) Value.
- 5) Velocity.

Big Data: Volume:

- * Related to an enormous size.
- * It is a vast 'volumes' of data generated from many source daily.



Veracity:

Structured, unstructured, semi-structured, that are being collected from different sources. Data will only be collected from datasets and sheets in the past.

Structured: It is in a tabular form, e.g: excel sheet, sql.

Unstructured: Rows and columns, Pixel

Semi-structured: JSON, XML, CSV, TSV and email. It stores in relation.

Veracity:

- * How much the data is reliable.
- * Manage data efficiently.
- * E.g: Facebook posts with hashtags.

Value:

- * Essential characteristics.
- * Not the data that we process or store.
- * It is valuable and reliable data that we store, process and analyze.

Velocity:

Attributes:

Qualities or characteristics that describe an object, individual or phenomenon.

1) Find all the duplicates in an array? For example: [1, 2, 5, 3, 1, 4, 5, 3, 1, 6, 1, 7, 2, 1]. solution [1, 2, 5, 3].

Input: 1, 2, 5, 3, 1, 4, 5, 3, 1, 6, 1, 7, 2, 1.

Freq. Table

$$\begin{aligned}\text{Mean} &= \frac{\sum x_i}{n} \\ &= \frac{14}{7} \\ &= 2.\end{aligned}$$

$I/P(n)$	$Freq(x)$
1	5
2	2
3	2
4	1
5	2
6	1
7	1

Types of Attributes: Qualitative Attributes and Quantitative Attributes.

Data Visualization:

It is a graphical representation of information and data.

Qualitative

Qualitative Attributes:

1) Nominal Attributes: Relating to names.

e.g: Skin color.

2) Binary Attributes: It is a category of nominal attributes that contains only two classes: 0 (or) 1.

3) Ordinal Attributes: It is a type of categorical data that possesses a meaningful order or ranking among its categories.

Quantitative Attributes:

1) Numeric Attributes: It is calculable, that is, it is a quantifiable amount that constitutes integer or value.

* Ratio - Scaled Attributes:

* Discrete Attributes: Has limited (or) restricted unlimited set of values which may appear as integer.

Basic Statistical Descriptions of Data:

1) Descriptive Statistics: Describe the basic features of data that provide a summary of the given data set which can either represent the entire population of a sample of the population.

1) Mean 2) Mode 3) Median.

↓
Arithmetic Average value that appears most often in a data set ↓
Middle value of sorted dataset.

1) Mean ($\mu/\bar{x}$): Arithmetic Average:

$$= \frac{\text{sum of values in dataset}}{\text{total no. of values}}$$

e.g: find the mean of the given dataset = {2, 4, 6, 8, 2, 10, 12}

Sol
$$\mu = \frac{2+4+6+8+2+10+12}{7} = \frac{44}{7} = 6.28$$

2) Find the mean value for dataset = {1, 2, 3, 4, 5, 11, 12, 13, 14, 15}

Sol
$$\mu = \frac{1+2+3+4+5+11+12+13+14+15}{10} = \frac{80}{10} = 8.$$

3) Median: The middle value of sorted data structure.

e.g: dataset = {1, 4, 2, 8, 2, 6, 10, 1, 12}

Sol Median = 4.
Sorted = {1, 1, 2, 2, 4, 6, 8, 10, 12}

e.g: 2: ds = {1, 4, 2, 8, 2, 6, 10, 1, 12, 5}

Sol: Sorted = {1, 1, 2, 2, 4, 5, 6, 8, 10, 12} = $\frac{4+5}{2} = 4.5$

Median: 4.5

For odd no. of ds: $M = \left[\frac{(n+1)}{2} \right]^{\text{th}}$ term.

For even no. of ds: $M = \left(\frac{n}{2} \right)^{\text{th}}$ term + $\left(\frac{n+1}{2} \right)^{\text{th}}$ term

Frequency Table:

The table contains the no. of occurrences of the data value is known as Frequency table.

e.g ds = [1, 2, 1, 3, 1, 2, 5, 6, 7, 5, 7]

Sol:

Input	freq
1	3
2	2
3	1
5	2
6	1
7	2

→ highest occurrence.

Mode: The most frequently occurred data in a given dataset.

e.g: From the above dataset value. After framing the frequency table. The mode is 1.

Python Program:

Mean:

ds = [1, 2, 3, 1, 3, 6, 5]

Output = [1, 3].

n = int(input())

arr = []

for i in range(n):

num = float(input())

arr.append(num)

mean = sum(arr)/n

print(mean)

Python Program:

Mean:

n = int(input())

arr = list(map(int, input().split()))

s = 0

for i in arr:

s += i

print(s/2)

Output:

1 2 4 5

6

Median:

Basic statistical descriptions:

Range: The difference between maximum and minimum value

Variance: (σ^2): It describes how much each number in a dataset differs from average, mean

e.g:

Calculate the variance of given dataset [75, 80, 85, 90, 95] as exam scores

Sol

Steps

$$\begin{aligned} 1) \text{ mean} &= \frac{75 + 80 + 85 + 90 + 95}{5} \\ &= \frac{425}{5} = 85 \end{aligned}$$

2) Calculate differences from mean as

$$\begin{aligned} &= (75 - 85), (80 - 85), (85 - 85), \\ &\quad (90 - 85), (95 - 85) \\ &= (-10), (-5), (0), (5), \\ &\quad (10) = [-10, -5, 0, 5, 10] \end{aligned}$$

3) Find the squares,

$$s^2 = [(-10)^2, (-5)^2, 0^2, 5^2, 10^2]$$

4) Calculate the mean of squared value.

$$\text{mean} = \frac{[100 + 25 + 0 + 25 + 100]}{5}$$

$$\sigma^2 = \frac{250}{5} = 50.$$

$$\text{standard deviation} = \sqrt{\sigma^2} = \sqrt{50} =$$

$$\boxed{7.07106}$$

so the standard deviation

Variance providing a more

There are 7 measures of shape and distribution as

- 1) Skewness Positive
negative
- 2) Kurtosis

→ mesokurtic
→ leptokurtic
→ plat kurtic

Data Visualization:

It is a graphical representation of 1) chart 2) graph 3) Map.

Tools: Cabul, Microsoft excel, Power BI.

Uses:

Unknown facts.

Create a path graph for python

Measuring Data Similarity:

Similarity Measuring:

It is a way of measuring how data samples are related to each other.

Similarity measuring:

How the data object are differ from one another

e.g: A(0.5, 0), B(1.5, 0), C(30, 0).



distance measure

⇒ A and B are similar.

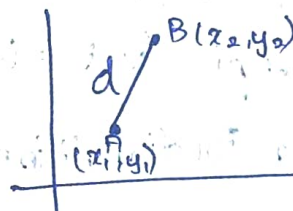
Distance

$$1) \text{ Euclidean dis} = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

$$2) \text{ Manhattan dis} = |y_2 - y_1|$$

↓
used in chess → bishop move

Distance Function:



$$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

Euclidean Distance.

It is the distance between two points A and B. To find euclidean points the length of the segment connecting the two points measured.

e.g find the distance between A(3, 2) and B(4, 1).

Sol: A(3, 2) B(4, 1)

$$\begin{aligned} d &= \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2} \\ &= \sqrt{(4 - 3)^2 + (1 - 2)^2} = \sqrt{1^2 + (-1)^2} \\ &= \sqrt{2} = 1.414. \end{aligned}$$

E.g. 2: A triangle has the vertices at points $A(2,3)$, $B(5,7)$, $C(8,1)$ find the length of the ~~the~~ longest side of the triangle.

Sol

1) find the distance between AB , BC , CA

2) Compare and find longest distance.

$$AB = \sqrt{(5-2)^2 + (7-3)^2} = \sqrt{3^2 + 4^2} \\ = \sqrt{9+16} = \sqrt{25} = 5$$

$$BC = \sqrt{(8-5)^2 + (1-7)^2} = \sqrt{3^2 + 6^2} \\ = \sqrt{9+36} = \sqrt{45} = 6.708$$

$$CA = \sqrt{(8-2)^2 + (1-3)^2} = \sqrt{6^2 + 2^2} \\ = \sqrt{36+4} = \sqrt{40} = 6.3$$

find greater length is $BC = 6.708$ units

Data mining and Functionalities

Data Mining: It is a technical method to ^{detect} ~~determine~~ information from huge dataset. There are two categories

1) Descriptive Data mining

It is used to summarize the data link as

- * Common relationship on data
- * Outliers

Techniques

- 1) clustering
- 2) Association rule mining

3) Visualization

2) Predictive Data Mining

It is consider with developing modules which can predict future behavior.

1) Decision trees

It is used to create a model that predict the value of target variable.

2) Neural Networks

E.g Speech recognition

The neural networks are used to create a model that can learn the packet recognition.

Regression Analysis

To create a model to predict the value of the target variable.

Data preparation Algorithms

It is a process of preparing a raw data to be suitable for

further processing and analysis. The raw data contains inconsistent, missing values, error and irrelevant informations.

Data preparation techniques like cleaning, imputation and normalization address this issues. Hence normalization

resize empty dataset.

Data preparation process

- 1) Describe the purpose
- 2) Data collection.

From variety to sources
~~and~~ files database and social media.

- 3) Data combining and integrating
- 4) Data profiling
- 5) Data exploring
- 6) Data transformation & enrichment
- 7) cleaning
- 8) Data validation.

Data visualization tool boxes for
~~data science~~ scientists

- 1) Apache Hadoop.
- 2) SAS - Statistical Analysis Systems
- 3) Apache Spark *
- 4) Data Robot
- 5) Tableau.
- 6) Big ML
- 7) TensorFlow
- 8) Jupyter.

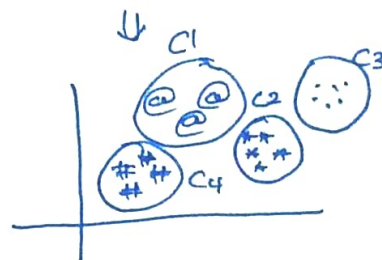
Unit - II

Clustering concepts:

Basics:



fig. dataset.



- => DSA MARK.
- * => CD MARK.
- # => SE MARK
- @ => MERN MARK
EVS

Clustering: It is known as grouping of same data from the given dataset. For grouping the distance between similar data is calculated for that some standard methods are used below:

1) Partitioning method.

* We use k-means algorithm.

k-means clustering: k is no. of clusters to be determine.

Algorithm: k-means

Input: K no. of clusters in which the dataset has to be ~~define~~ divided. The dataset is divided as d. cluster center is needed for n number of Objects.

Output: Dissimilar dataset of k clusters

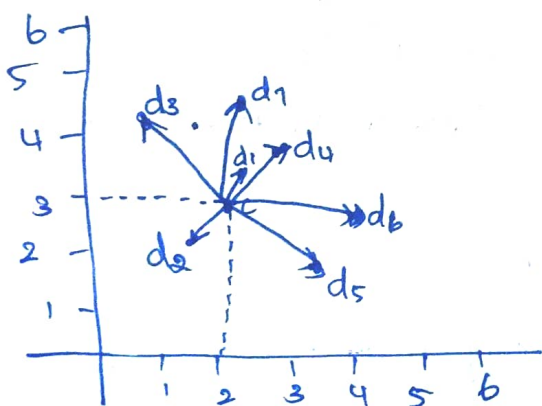
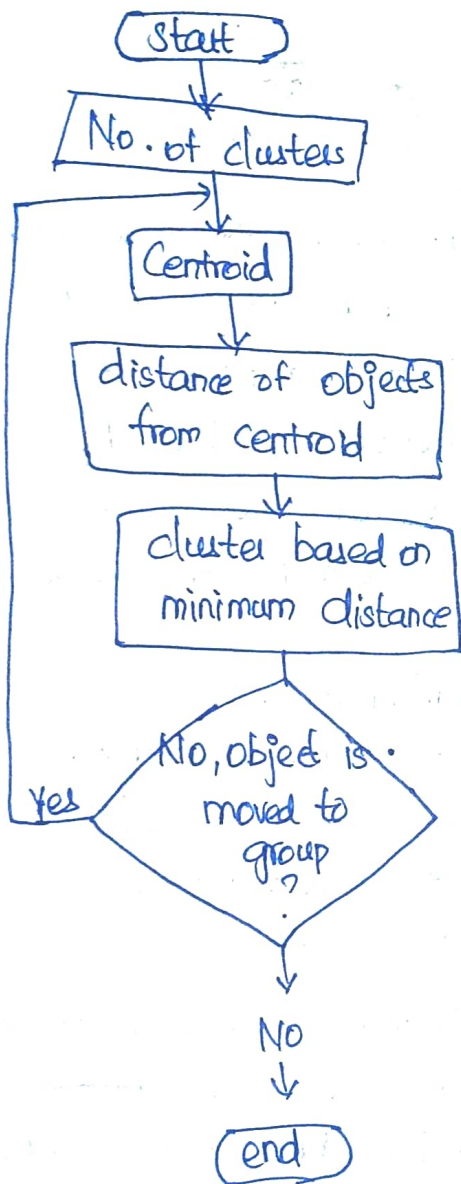
Algorithm:

- 1) Randomly assign k objects from

the dataset.

- 2) Assign each object to the object which is the most similar based on mean values.
- 3) Update the mean value
- 4) Repeat step 2 until no change

Flowchart:



16, 16, 17, 20, 20, 21, 21, 22, 23
29, 36, 41, 42, 43, 44, 45, 61, 6

Initial clusters:

$k=2$

Randomly choose 2 centroids from the given dataset ~~centroid~~

Centroid $c_1 = 16$.

Centroid $c_2 = 22$.

Iteration 1

For c_1 , find the mean by adding the nearest values 16, 16, 17

$$\therefore c_1 = (16 + 16 + 17) / 3 = 16.33$$

The remaining is added and determine mean as

$$c_2 = [20 + 20 + 21 + 21 + 22 + 23 + 29 + 36 + 41 + 42 + 43 + 44 + 45 + 61 + 66] / 16 = 37.25$$

Iteration 2

$$c_1 = [16 + 16 + 17 + 20 + 20 + 21 + 21 + 22 + 23] / 9 = 19.55$$

$$c_2 = [29 + 36 + 41 + 42 + 43 + 44 + 45 + 61 + 62 + 66] / 10 = 46.90$$

Iteration - 3 :

$$c_1 = [16 + 16 + 17 + 20 + 20 + 21 + 21 + 23 + 22 + 29] / 10 = 20.50$$

$$c_2 = [36 + 41 + 42 + 43 + 44 + 45 + 61 + 62 + 66] / 9 = 48.89$$

Iteration - 4

$$C_1 = \frac{10+16+17+20+20+21+21+22+23+29}{10} = 20.50$$

Since there is no move

$$C_2 = 48.89$$

* No change between Iteration 3 and 4.

* So we set the clusters (16-29) and (36-66).

Calculate: 1.1

Iteration 1.1

$$C_1 = \frac{16+16+17+20+20}{5} = \frac{89}{5} = 17.8$$

$$C_2 = \frac{21+21+22+23+29+36+41+42+43+44+45+61+62+66}{14}$$

$$= \frac{556}{14} = 39.71$$

② dataset = {10, 10, 11, 11, 12, 13, 14, 15, 16, 17, 20, 21, 22, 24, 26, 32, 33}

k=2.

$$C_1 = 10 \quad C_2 = 12$$

Iteration 1

$$\therefore C_1 = \frac{10+10+11+11+12}{5} = 10.8$$

$$C_2 = \frac{13+14+15+16+17+20+21+22+24+26+32+33}{12}$$

$$= \frac{253}{12} = 21.083$$

Iteration: 2

$$C_1 = \frac{10+10+11+11+12+13+14+15}{8} = \frac{96}{8} = 12$$

$$C_2 = \frac{16+17+20+21+22+24+26+32+33}{9} = \frac{241}{9} = 26.77$$

Iteration: 3

$$C_1 = \frac{10+10+11+11+12+13+14+15+16}{9} = \frac{112}{9} = 12.44$$

$$C_2 = \frac{17+20+21+22+24+26+32+33}{8} = \frac{195}{8} = 24.37$$

21/06/26

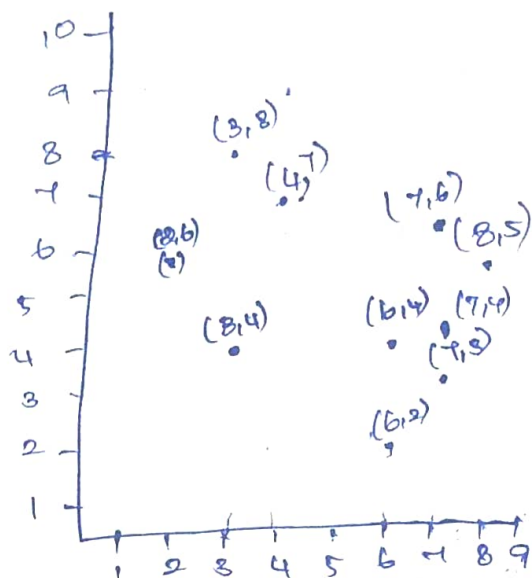
K-medoids

It is a unsupervised learning cluster algorithm that partition an amount of k-distinct group by using a actual data points (medoids) as cluster centers.

Unlike k-means, which calculates an artificial mean for each cluster, the k-medoids ensures that the center of a cluster is always an existing object from the database.

Q.9

i	x	y	cen 1	cen 2	cen 3
x1	2	6	3	7	C1
x2	3	4	0	4	C1
x3	3	8	4	8	C1
x4	4	7	4	6	C1
x5	6	2	5	3	C2
x6	6	4	3	1	C2
x7	7	3	5	1	C2
x8	7	4	4	0	C2
x9	8	5	6	2	C2
x10	7	6	6	2	C2



Steps:

1. Randomly select 2 medoids

(i.e) $C_1 = (3, 4)$

$C_2 = (7, 4)$

2) Use manhattan distance.

$(x_1, y_1) (x_2, y_2)$
 $(3, 4) (8, 6)$

midistance = $|x_2 - x_1| + |y_2 - y_1|$

$= |2-3| + |6-4| = 1+2=3$

Hierarchical Method

In this method, grouping data into three of clusters.

Steps:

1) Identify the two clusters which can be closer together.

2) Met maximum two comparable clusters (The method starts by treating each data point as a separate cluster).

3) If iteratively combine the closest cluster until the stopping point (or) condition is reached.

A) The final result like a tree structure and it is called dendrogram

Types:

1) Agglomerative clustering

In this method consider every data point as an individual cluster and met nearest cluster points.

Algorithm:

1) Calculate the similarity of one cluster with all other clusters. (calculate the proximity matrix).

2) consider every point as a individual cluster.

3) Merge the clusters.

4) Recalculate the proximity matrix.

5) Repeat step 3 & 4 until a single cluster is obtain are reach.

Ex: Consider the dataset

{18, 22, 25, 27, 42, 43} Apply Agglomerative clustering concept and find the cluster to build the dendrogram.

Sol

The given dataset = {18, 22, 25, 27, 42, 43} find the minimum distance

1) Merge the cluster using minimum distance and update the proximity matrix accordingly.

1	18	22	25	27	42	43
18	0	4	7	9	24	25
22	4	0	3	5	20	21
25	7	3	0	2	17	18
27	9	5	2	0	15	16
42	24	20	17	15	0	1
43	25	21	18	16	1	0

2) Find the minimum distance of the table and met those two values

	18	22	25	27	42, 43	43
18	0	4	7	9	24	
22	4	0	3	5	20	
25	7	3	0	2	17	
27	9	5	2	0	15	
42, 43	24	20	17	15	0	
43						

3) Find the minimum distance again and now it is found at the values 25, 27. so remove the values of 27th column and 27 row and merge it into 25

1	18	22	25, 27	42, 43
18	0	4	7	24
22	4	0	3	20
25, 27	7	3	0	17
42, 43	24	20	17	0

The merged clusters are {42, 43}, {25, 27}

4) Now, find the next minimum distance.

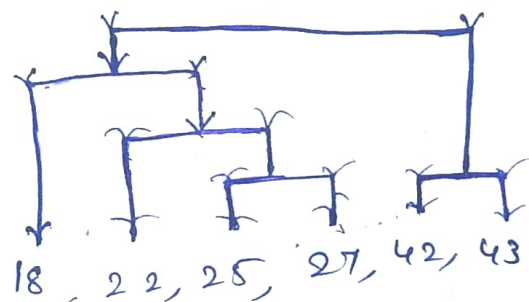
	18	22, 25, 27	42, 43
18	0	4	24
22, 25, 27	4	0	20
42, 43	24	20	0

Now we get merge 18 also.

Now we get {42, 43} (~~18, 22, 25, 27~~)

((22, 25, 27) 22) 18

Now we draw the dendrogram.



Density Based clustering

It is referred as one of the unsupervised learning method. The data points are separated by two clusters. The points to be consider as

1) Eps - Maximum radius of the neighbourhood.

2) Minpts - Refers to minimum number of points.

DBSCAN $\Rightarrow$ Density Based Spatial Clustering of Applications with Noise $\rightarrow$ error, deviation.

It depends on density based evaluation of cluster. It identifies the clusters of orbitally sides in the spatial database with outliers.

Eg: 1, 2, 3 ... 10 to be added

In the evaluation instead of 10 there is 100 it is called outlier.

DBSCAN clustering:

e.g Apply ^{DBSCAN} and find the cluster.

Datapoints: $P_1(3,7), P_2(4,6)$.

$P_3(5,5), P_4(6,4), P_5(7,3), P_6(6,2),$
 $P_7(7,2), P_8(8,4), P_9(3,3), P_{10}(2,6)$
 $P_{11}(3,5)$ and $P_{12}(2,4)$

Given: minpts = 4 & Epsilon(ϵ) = 1.9
to create clusters.

Sol
Given: minpts = 4
 $\epsilon = 1.9$

The minpts refers to number of datapoints that a cluster should contain. Here minimum of 4 data points, that a cluster name have points. The minimum distance between each data point is less than or equal to 4. Here the minimum distance ϵ is 1.9. To calculate the distance between each data points, is done with euclidean distance.

1) The minpts = 4 & $\epsilon = 1.9$

The distance is calculated as P_1 to $P_1 = 0$.

$P_1(3,7)$ to $P_2(4,6) = \sqrt{(4-3)^2 + (6-7)^2} = \sqrt{2} = 1.4$

$P_1(3,7)$ to $P_3(5,5) = \sqrt{(5-3)^2 + (5-7)^2} = \sqrt{8} = 2.82$

$P_6(6,2)$ to $P_1(3,7) = \sqrt{(5-6)^2 + (7-2)^2} = \sqrt{34} = 5.8$

We have to identify nearest point

Horizontally identify the distance.

$\epsilon \leq 1.9$.

Find the nearest points and

find the noise.

For example for P_1 the

nearest points are P_1, P_2, P_{10}

where it is less than the 4 number of minimum points 4.

That is the minimum points must be 4. The condition is not satisfied as it should have 4 minimum points. It is called noise.

	P_1	P_2	P_3	P_4	P_5	P_6	P_7	P_8	P_9	P_{10}	P_{11}	P_{12}
P_1	0											
P_2	1.41	0										
P_3	2.83	1.41	0									
P_4	4.24	2.83	1.41	0								
P_5	5.6	4.24	2.83	1.41	0							
P_6	5.83	4.47	3.16	2	1.41	0						
P_7	6.4	5	3.61	2.24	1	1	0					
P_8	5.83	4.47	3.16	2	1.41	2.83	2.24	0				
P_9	4	3.16	2.83	3.16	4	3.16	4.12	5.10	0			
P_{10}	1.41	2	3.16	4.47	5.83	5.66	6.4	6.32	3.16	0		
P_{11}	2	1.41	2	3.16	4.47	4.24	5	5.10	2	1.41	0	
P_{12}	3.16	2.83	3.16	4	5.10	4.47	5.39	6	1.41	2	1.41	0

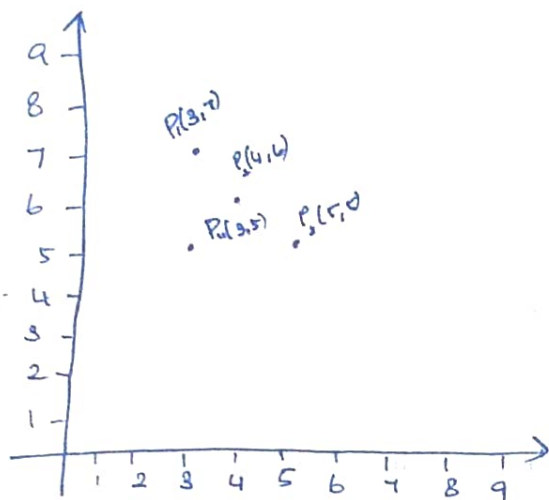
nearest point:

$P_1: P_2, P_{10}$
 $P_2: P_3, P_{11}, P_1$
 $P_3: P_4, P_5$
 $P_4: P_5, P_6$
 $P_5: P_4, P_6, P_7, P_8$
 $P_6: P_5, P_7$
 $P_7: P_5, P_6$
 $P_8: P_5$
 $P_9: P_{12}$
 $P_{10}: P_1, P_{11}$
 $P_{11}: P_2, P_{10}, P_{12}$
 $P_{12}: P_9, P_{11}$

Minpoints = 4

Points	status	
P_1	noise	Border
P_2	core	P
P_3	noise	Border
P_4	noise	Border
P_5	core	
P_6	noise	Border
P_7	noise	Border
P_8	noise	
P_9	noise	
P_{10}	noise	Border
P_{11}	core	
P_{12}	noise	Border

Considering the core points P_2, P_5, P_{11} these are defined as cluster center points.



Taking P_2, P_5, P_{11} are the cluster centers
 Taking P_2 as the center point
 Consider P_5 as the next center point
 P_4, P_6, P_7 and P_8 are the nearest points to be P_5 . So they are grouped together to form a cluster. 2.

DENCLUE

Density Based Clustering

$$\hat{f}_b(x) = \frac{1}{nh} \sum_{i=1}^n k\left(\frac{x - x_i}{h}\right)$$

kernel function

The probability density at a point depends on the distance from that point to the observed point is determined using the above formula.

The density function is based on observed data.

The object into finite no. of clusters. The multi resolution is used to process quickly the grid data structure. There are two types.

1) Statistical data stored in grid size (STING).

2) Defines a grid and density based approach.

~~Where else clustering using hierarchies~~

Birch Clustering Algorithm

Balanced iterating reducing and clustering hierarchies

It is an unsupervised algorithm:

*) Consider the threshold value

*) For each data we need to evaluate radius and cluster features. The cluster features are known as $\langle N, LS, SS \rangle$

where N is no. of data points.

LS is sum of data point

SS is squared sum of data point

For each data point we need to calculate radius.

$$\text{Radius} = \sqrt{\frac{SS - \frac{LS^2}{N}}{N}}$$

$x_1(3, 4), x_2(2, 6), x_3(4, 5), x_4(4, 7), x_5(3, 8), x_6(6, 2), x_7(7, 2), x_8(7, 4), x_9(8, 4), x_{10}(7, 9)$, Threshold $T \leq 1$ and max branch is 2.

Sol ~~take $x_1 = 3, 4$~~

Step 1: take $x_1 = 3, 4$

$L_S, SS - NA$

Radius = 0 (one value)

$N=1$, $L_S = (3, 4)$, $SS = (3^2, 4^2)$

$CF_1 < 1, (3, 4), (9, 16)$

create the leaf node

$CF_1 < 1, (3, 4), (9, 16)$

leaf
(3, 4)

Step 2:

$x_1 (3, 4)$, $x_2 (2, 6)$

$N=2$ $L_S = ((3+2), (4+6)) = (5, 10)$

$SS = \cancel{SS=100} (3^2+2^2), (4^2+6^2) = (13, 52)$

$$\text{Radius} = \sqrt{\frac{(13, 52) - \frac{(5^2, 10^2)}{2}}{2}}$$

$$= \sqrt{\frac{(13, 52) - (12.5, 50)}{2}}$$

$$= \sqrt{\frac{(3, 52), (52-50)}{2}}$$

$$= \sqrt{\frac{0.5, 2}{2}} = \sqrt{0.25, 1} < 1.5$$

$CF_2 < 2, (5, 10), (13, 52)$

leaf
(3, 4)
(2, 6)

Step 3: $x_3 (4, 5)$

$N=3$

$L_S = ((5+4), (10+5)) = (9, 15)$

$SS = \cancel{((5^2+4^2), (10^2+5^2)) = (41, 125)}$

$SS = ((3+4^2), (52+5^2)) = (29, 77)$

$$\text{Radius} = \sqrt{\frac{(29, 77) - \frac{(9, 15)^2}{3}}{3}}$$

$$= 0.47, 0.4714 < 1.5$$

$CF_1 < 3, (9, 15), (29, 77)$

leaf
 $x_1 = 3, 4$
 $x_2 = 2, 6$
 $x_3 = 4, 5$

$x_4 = 4, 7$

$N=4$, $L_S = (9, 15) + (4, 7) = (13, 22)$

$SS = (29, 77) + (4^2 + 7^2)$
 $= (45, 126)$

$$R = \sqrt{\frac{(45, 126) - \frac{(13, 22)^2}{4}}{4}}$$

$$= \sqrt{\frac{(45, 126) - (169, 484)}{4}}$$

$$= \sqrt{(145-169), (126-484)}$$

$$= (0.41, 0.55) < 1.5 \text{ (True)}$$

$CF_1 < 4, (13, 22), (45, 126)$

leaf
 $x_1 = 3, 4$
 $x_2 = (2, 6)$
 $x_3 = 4, 5$
 $x_4 = 4, 7$

$$x_5 = 3,8$$

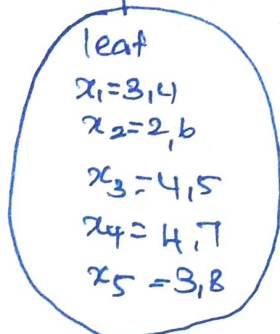
$$N = 5$$

$$LS = 16,30$$

$$SS = 54,190$$

$$R = 0.33, 0.63 < 1 = \text{True}$$

$$CF_1 < 5(16,30), (54,190)$$



$$x_6 = 6,2$$

$$LS = 22,32, SS = 90,194$$

$$N = 6$$

$$\text{Radius} = \sqrt{\frac{(22,32) - (90,194)^2}{6}}$$

$$= \sqrt{\frac{(22,32) - (8100, 37,638)}{6}}$$

$$\text{Radius} = \sqrt{\frac{(90,194) - \frac{(22^2, 32^2)}{6}}{6}}$$

$$= \sqrt{\frac{(90,194) - \frac{(484, 1024)}{6}}{6}}$$

$$= \sqrt{\frac{(90,194) - (80.67, 170.67)}{6}}$$

$$= \sqrt{\frac{(90 - 80.67), (194 - 170.67)}{6}}$$

$$= \sqrt{\frac{9.33, 23.33}{6}} = \sqrt{1.55, 3.89}$$

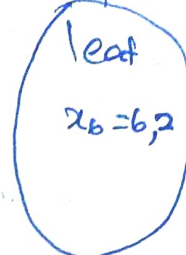
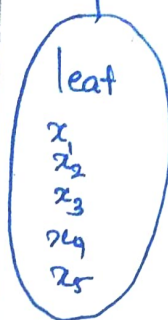
$$= 1.25, 1.97$$

$$= 0.508, 0.805 < 1.5$$

$$= 1.24, 1.97 < 1.5 = \text{false}$$

$$CF_1 < 5(16,30), (54,190)$$

$$CF_2 < 6, (22,32), (90,194)$$



$$CF_2 < N, LS, SS$$

$$N = 1, x_6 = 6,2, x_7 = (7,2)$$

$$LS = (6,2) + (7,2) = (13,4)$$

$$SS = (6^2 + 7^2, 2^2 + 2^2)$$

$$= (85, 8)$$

$$\text{Radius} = \sqrt{\frac{(85,8) - \frac{(13,4)^2}{2}}{2}}$$

$$= \sqrt{\frac{(85,8) - (169,16)}{2}}$$

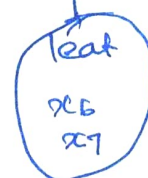
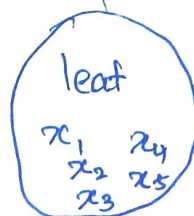
$$= \sqrt{\frac{(85,8) - (84.5, 4)}{2}}$$

$$= \sqrt{\frac{0.5, 4}{2}} = \frac{0.507, 2}{2}$$

$$= (0.3, 1) < 1.5$$

$$CF_1 < 5(16,30), (54,190)$$

$$CF_2 < 2, (13,4), (85,8)$$



$$x_8 = 7,4 \quad N = 8$$

$$LS = (20,8)$$

$$SS = (134,24)$$

$$R = \sqrt{\frac{SS - \frac{LS^2}{N}}{N}} = \sqrt{\frac{(134,24) - \frac{(20,8)^2}{8}}{8}}$$

~~0.7,1~~

$$= \sqrt{\frac{(134,24) - (100,64)}{8}}$$

$$= \sqrt{\frac{(134,24) - (80,8)}{8}}$$

$$= \sqrt{\frac{(134,24) - (133,34,21,34)}{8}}$$

$$= \sqrt{\frac{(134 - 133,34), (24 - 21,34)}{8}}$$

$$= \sqrt{\frac{0,66, 2,66}{8}} = \sqrt{0,22, 0,886}$$

$$= 0,47, 0,94 < 1,5$$

$\therefore R < T(1,5)$ [True]

$$CF1 < 5, (16,30), (54,90)$$

$$CF2 < 3, (20,8), (134,24)$$

leaf

$$x_1 = 3,4$$

$$x_2 = 2,6$$

$$x_3 = 4,5$$

$$x_4 = 4,7$$

$$x_5 = 3,8$$

leaf

$$x_6 = 6,2$$

$$x_7 = 7,2$$

$$x_8 = 7,4$$

$$x = 9, N = 4$$

$$LS = (28,12)$$

$$SS = (198,40)$$

$$R = \sqrt{\frac{(198,40) - (28,12)^2}{4}}$$

$$= \sqrt{\frac{198,40 - (184,144)}{4}}$$

$$= \sqrt{\frac{198,40 - 196,36}{4}}$$

$$= \sqrt{\frac{(198-196), (40-36)}{4}}$$

$$= \sqrt{\frac{2,4}{4}} = \sqrt{0,5,1}$$

$$= 0,7,1 < 1,5$$

$\therefore R < T(1,5)$ [True]

$$CF1 < 5, (16,30), (54,90)$$

$$CF2 < 4, (28,12), (198,40)$$

leaf

$$x_1 = 3,4$$

$$x_2 = 2,6$$

$$x_3 = 4,5$$

$$x_4 = 4,7$$

$$x_5 = 3,8$$

leaf

$$x_6 = 6,2$$

$$x_7 = 7,2$$

$$x_8 = 7,4$$

$$x_9 = 8,4$$

Step 10:

$$x_{10} = (7,9) \quad N = 5$$

$$SS = (247,121) \quad LS = 35,21$$

$$R = \sqrt{\frac{(247,121) - \frac{(35,21)^2}{5}}{5}}$$

$$= 0,63, 2,56 < 1,5$$

$\therefore R < T$ (False)

~~CF1 < 5~~

$$CF1,2 < 9, (44,42), (202,230)$$

$$CF3 < 1, (7,9), (49,81)$$

$$CF1 < 5, (16,30), (54,90)$$

$$CF2 < 4, (28,12), (198,40)$$

leaf

$$x_1 = 3,4$$

$$x_2 = 2,6$$

$$x_3 = 4,5$$

$$x_4 = 4,7$$

$$x_5 = 3,8$$

leaf

$$x_6 = 6,2$$

$$x_7 = 7,2$$

$$x_8 = 7,10$$

$$x_9 = 8,4$$

leaf

$$x_{10} = 7,9$$

x_0 does not belong to CF_2 so create one more leaf node and add it now check how many maximum branches allowed since it is indicated as 2, we cannot add any further branch or leaf node. Merge CF_1 and CF_2

and store it in CF_2 . Then create CF_3 . that is $CF_2 = (CF_1 < 5, (16, 30), (54, 10)) + (CF_2 < 1, (28, 12), (198, 40))$. The new value for $CF_2 < 9, (144, 12), (252, 230)$
 $CF_3 < 1, (7, 9), (149, 81)$

$CF_2 < 9, (144, 12), (252, 230)$
 $CF_3 < 1, (7, 9), (149, 81)$

$CF_1 < 5, (16, 30), (54, 10)$

UNIT-III $\Rightarrow$ Classification.

Data classification:

* Process of classifying data in relevant categories

Introduction

* Classification is a method that enables machines to organize items into specific groups.

* It learns by analyzing labeled examples - such as email tagged as "spam" or "not spam".

* Once trained, the system can categorize new data, like determining whether a

fresh email is open.

KNN - K-nearest Neighbour algo

Types of classification

- * Supervised learning algorithm
- * Unsupervised learning algorithm

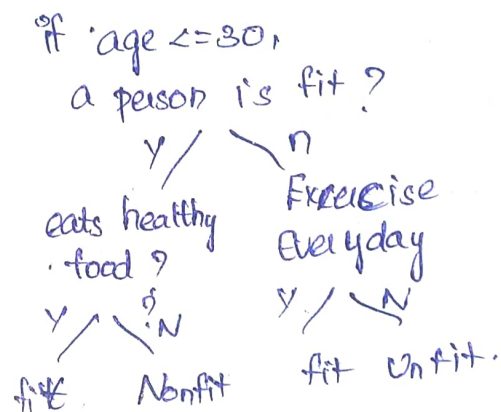
Classification types:

1) Binary classification.

- * The goal is to sort the data into two distinct categories.
- * Sort the emails spam or not.

<https://www.youtube.com/watch?v=01VwW8T1Kcg>

Decision Trees:



Here, a person is determined

There are two attribute selection measures for you.

1) Information gain: explain the estimate uncertain.

It follows two steps:

- 1) Entropy
- 2) Information gain

2) Gini Index:

It is a tree consist of ~~two~~ node as attribute and the branches of outcome.

2.9

Day	Outlook	Temp	Humidity	Wind	Play Games
				Week	No.
D ₁	Summer	Hot	High	Week	No
D ₂	Summer	Hot	High	Strong	Yes
D ₃	Over cast	Hot	High	Week	Yes
D ₄	Rain	Mild	High	Week	Yes
D ₅	Rain	Cool	Normal	Week	No.
D ₆	Rain	cool	Normal	Week Strong	Yes
D ₇	Over cast	cool	Normal	Strong	No
D ₈	Summer	Mild	High	Week	Yes
D ₉	Summer	cool	Normal	Week	Yes
D ₁₀	Rain	Mild	Normal	Week	Yes
D ₁₁	Summer	Mild	Normal	Strong	Yes
D ₁₂	Overcast	Mild	High	Strong	Yes
D ₁₃	overcast	Hot	Normal	Week	Yes
D ₁₄	Rain	Mild	High	Strong	No

Based on the attribute ^{a decision is made} whether

a player play games (or) not.

Sol:

No. of attributes = 4.

Take a outlook attribute calculate the information gain of the maximum attribute.

We need to calculate entropy of entire dataset. and individual data like Summer, ~~over~~ overcast, Rain attributes where entropy is known as the measure of disorder.

$$\text{Entropy}(S) = -p(\text{yes}) \log_2 p(\text{yes}) - p(\text{no}) \log_2 p(\text{no})$$

$$\text{Entropy}(S) = \text{Positive} + \text{negative} \\ 9 + 5 = 14.$$

$$\begin{aligned} E(S) &= -\frac{9}{14} \log_2 \left(\frac{9}{14}\right) - \frac{5}{14} \log_2 \left(\frac{5}{14}\right) \\ &= -\frac{9}{14} [\log_2(9) - \log_2(14)] - \frac{5}{14} [\log_2(5) - \log_2(14)] \\ &= \cancel{0.405} - \frac{9}{14} [3.17 - 3.80] - \frac{5}{14} [2.322 - 3.80] \\ &= -\frac{9}{14} (-0.63) - \frac{5}{14} (-1.478) \\ &= 0.405 - 0.1802 \\ &= 0.94 \end{aligned}$$

~~The entropy of~~

$$\text{Entropy}(S_{\text{summer}}) = [2(y), 3(n)]$$

$$E(\text{summer}) = -\frac{2}{5} \log_2 \left(\frac{2}{5}\right) - \frac{3}{5} \log_2 \left(\frac{3}{5}\right)$$

$$= -\frac{2}{5} [\log_2(2) - \log_2(5)] - \frac{2}{5}$$

$$= [\log_2(3) - \log_2(5)]$$

$$0.971$$

$$2) \text{Entropy (Overcast)} \leftarrow [4, 0]$$

$$= \frac{4}{4} \log_2(4/4) - \frac{0}{4} \log_2(0/4)$$

$$= 0 - 0 = 0.$$

If only possible (or) only negative is that the entropy is treated as 0.

$$3) \text{Entropy (rain)} \leftarrow [3, 2] \rightarrow 5$$

$$= \frac{3}{5} \log_2(3/5) - \frac{2}{5} \log_2(2/5)$$

$$= 0.971.$$

Gain(s, Outlook)

↓
dataset.

$$\text{Entropy}(s) - \sum \frac{|S_v|}{|S|} \text{Entropy}(s_v)$$

$$\text{Entropy}(s_v)$$

$$= \text{Entropy}(s) - \frac{5}{14} \text{Entropy}(\text{Summer}) -$$

$$\frac{4}{14} \text{Entropy}(\text{Overcast}) -$$

$$\frac{5}{14} \text{Entropy}(\text{rain})$$

$$= 0.94 - \frac{5}{14}(0.971) - \frac{4}{14}(0) - \frac{5}{14}(0.971)$$

$$= 0.94 - \frac{5}{14}(0.971) - 0 - \frac{5}{14}(0.971)$$

$$= 0.94 - (0.3511)(0.971) - 0 - (0.3511)(0.971)$$

$$= 0.2464.$$

Similarly we need to calculate attribute. the entropy of the attributes temp, Humidity, wind.

Consider the output as a root node.

⇒ It has 3 values.

Summer	Overcast	rain
D ₁	D ₃	D ₄
D ₂	D ₇	D ₅
D ₈	D ₁₂	D ₆
D ₉	D ₁₃	D ₁₀
D ₁₁		D ₁₄

Hence the decision tree makes

{D₁, D₂, D₃, ..., D₁₄}

[9(yes), 5(No)]

Outlook

Summer	Overcast	rain
{D ₁ , D ₂ , D ₃ , D ₉ , D ₁₁ }	{D ₃ , D ₇ , D ₁₂ , D ₁₃ }	{D ₄ , D ₅ , D ₆ , D ₁₀ , D ₁₄ }
[2 yes, 3 No]	[4 yes, 0 no]	[3 yes, 2 No]
[yes/no]	[Yes]	[Yes/No]

Hence considering a single output value either yes (or) no. Here where the player can play a game.

KNN - k - Nearest Neighbour Algorithm
Lazy-learn algorithm:

→ Supervised.

- k - closest point.

To make the decision based on the average value. It classifies data based on similarity with nearby data points. Here, the distance between ~~based on euclidean~~ between the nearest points using euclidean distance.

KNN is also called lazy learning algorithm because it does not learn from the training set immediately. Instead it stores the entire dataset and performs computations only at a time of classification. The k is known as the number of nearest points.

Using
1) O Method (or)

2) Statistical Method (or)

Randomly fix odd values to k .

e.g. Evaluate the nearest mark values to determine whether the student is pass?

dataset:

S.No	m	cs	Result
1	4	3	Fail
2	6	7	Pass
3	7	8	Pass
4	5	5	Fail
5	8	8	Pass

Determine the 6th Values evaluated as pass or fail.

Sol

dataset : Step 1: Finding distance.

S.No	m	cs	Result	distance
1	4	3	Fail	5.39
2	6	7	Pass	1
3	7	8	Pass	1
4	5	5	Fail	3.16
5	8	8	Pass	2

take a random point 6, 8 for m , cs respectively.

$$x_2 = 6, y_2 = 8.$$

$$E_d = \sqrt{(6-4)^2 + (8-3)^2} = \sqrt{2^2 + 5^2} = \sqrt{4+25} = \sqrt{29} = 5.38$$

$$E_{d2} = \sqrt{(6-6)^2 + (8-7)^2} = 1.$$

~~Step 1~~
Step 2: Arrange the values in ascending order. based on distance

m	cs	Result	Distance
6	7	Pass	1
7	8	Pass	1
8	8	Pass	2
5	5	Fail	3.16
4	3	Fail	5.39

Step 3: fix $k=3$.

m	cs	Result	distance
6	7	P	1
7	8	P	1
8	8	P	2
6	8	P	0

So the majority is based on the mark is pass, so the last value is also derived as pass.

Entropy (s) = $-p(\text{yes}) \log_2 p(\text{yes}) - p(\text{no}) \log_2 p(\text{no})$

$E(s) = 0.94$
Entropy (wind) = Positive + negative = $6+2=8$.

Entropy (cool) = $-\frac{6}{8} \log_2 \left(\frac{6}{8}\right) - \frac{2}{8} \log_2 \left(\frac{2}{8}\right)$

Entropy (cool) = $[3,1]$

$E(\text{cool}) = -\frac{3}{4} \log_2 \frac{3}{4} - \frac{1}{4} \log_2 \frac{1}{4}$

$$\begin{aligned}
 &= \frac{3}{4} (\log_2 3 - \log_2 4) - \\
 &\quad \frac{1}{4} (\log_2 (1) - \log_2 4) \\
 &= \frac{3}{4} [1.585 - 2] - \\
 &\quad \frac{1}{4} [0 - 2] = \frac{3}{4} [-0.415] - \\
 &\quad \frac{1}{4} [-2] = 0.811
 \end{aligned}$$

$$\text{Entropy (Hot)} \leftarrow [2, 2]$$

$$\begin{aligned}
 f(\text{Hot}) &= -\frac{2}{4} \log_2 \frac{2}{4} - \frac{2}{4} \log_2 \frac{2}{4} \\
 &= -\frac{1}{2} (-1) - \frac{1}{2} (-1) \\
 &= 0.5 + 0.5 = 1
 \end{aligned}$$

$$\text{Entropy (Mild)} \leftarrow [4, 2]$$

$$\begin{aligned}
 f(\text{Mild}) &= -\frac{4}{6} \log_2 \frac{4}{6} - \frac{2}{6} \log_2 \frac{2}{6} \\
 &= -\frac{2}{3} \log_2 \frac{2}{3} - \frac{1}{3} \log_2 \frac{1}{3} \\
 &= -\frac{2}{3} (\log_2 (2) - \log_2 (3)) - \\
 &\quad -\frac{1}{3} (\log_2 (1) - \log_2 (3)) \\
 &= -\frac{2}{3} (1 - 1.585) - \\
 &\quad -\frac{1}{3} (0 - 1.585) \\
 &= 0.918
 \end{aligned}$$

Gain(s, Temp)

$$\text{Entropy (s)} = \sum \frac{(\Delta V)}{\Delta I}$$

$$\begin{aligned}
 &= \text{Entropy (s)} - \frac{H}{11} \text{Entropy (hot)} - \\
 &\quad \frac{6}{11} \text{Entropy (mild)} - \frac{4}{11} \text{Entropy (cool)} \\
 &= 0.94 - \frac{H}{11} (1) - \frac{6}{11} (0.918) - \\
 &\quad \frac{4}{11} (0.811) \\
 &= 0.94 - 0.286 - 0.393 - 0.292 \\
 &= 0.059
 \end{aligned}$$

3) Entropy

Hot	Mild	cool
D1	D4	D5
D2	D8	D6
D3	D10	D7
D9	D11	D9
	D12	
	D14	

Hot	Mild	cool
{D1, D2, D3, D9}	{D4, D8, D10, D11, D12}	{D5, D6, D7, D9}
2 yes, 2 no.	4 yes, 2 no.	3 yes, 1 no.

Humidity

$$\text{Entropy (High)} \leftarrow [3, 4]$$

$$\begin{aligned}
 &= -\frac{3}{7} \log_2 \frac{3}{7} - \frac{4}{7} \log_2 \frac{4}{7} \\
 &= -\frac{3}{7} (\log_2 (3) - \log_2 (7)) - \\
 &\quad \frac{4}{7} (\log_2 (4) - \log_2 (7)) \\
 &= -\frac{3}{7} (1.585 - 2.807) - \\
 &\quad \frac{4}{7} (2 - 2.807) \\
 &= -\frac{3}{7} (-1.222) - \frac{4}{7} (-0.807) \\
 &= 0.523 + 0.461 \\
 &= 0.985
 \end{aligned}$$

$$\text{Entropy (Normal)} \leftarrow [6, 1]$$

$$\begin{aligned}
 &= -\frac{6}{7} \log_2 \frac{6}{7} - \frac{1}{7} \log_2 \frac{1}{7} \\
 &= -\frac{6}{7} (2.585 - 2.807) - \frac{1}{7} (0 - 2.807) \\
 &= -\frac{6}{7} (-0.222) - \frac{1}{7} (-2.807) \\
 &= 0.190 + 0.401 \\
 &= 0.592
 \end{aligned}$$

Gain(S, Humidity)

$$\begin{aligned}
 & \text{Entropy}(S) - \frac{7}{14} \text{Entropy}(\text{High}) - \frac{7}{14} \text{Entropy}(\text{Normal}) \\
 &= 0.94 - \frac{7}{14} (0.985) - \frac{7}{14} (0.592) \\
 &= 0.94 - 0.4925 - 0.296 \\
 &= 0.1515
 \end{aligned}$$

Humidity

High	Normal
{D ₁ , D ₂ , D ₃ , D ₄ , D ₆ , D ₁₂ , D ₁₄ }	{D ₅ , D ₆ , D ₇ , D ₉ , D ₁₀ , D ₁₁ , D ₁₃ }
[3 Yes, 4 No]	[6 Yes, 1 No]
No	Yes

For Wind.

$$\text{Entropy}(S) = 0.94$$

$$\text{Entropy}(\text{weak}) = [6, 2]$$

$$= -\frac{6}{8} (\log_2 6 - \log_2 8) -$$

$$\frac{2}{8} (\log_2 2 - \log_2 8)$$

$$= -\frac{6}{8} (2.585 - 3) -$$

$$\frac{2}{8} (1 - 3)$$

$$= -\frac{6}{8} (-0.415) - \frac{2}{8} (-2)$$

$$= 0.31 + 0.500 = 0.811$$

$$\text{Entropy}(\text{strong}) = [3, 3]$$

$$E(\text{strong}) = -\frac{3}{6} \log_2 \frac{3}{6} - \frac{3}{6} \log_2 \frac{3}{6}$$

$$= -\frac{1}{2} \log_2 \frac{1}{2} - \frac{1}{2} \log_2 \frac{1}{2}$$

$$= -\frac{1}{2} (-1) - \frac{1}{2} (-1)$$

$$= 0.5 + 0.5 = 1$$

$$\text{Gain}(S, \text{Wind}) = 0.94 - \frac{8}{14} (0.811) - \frac{6}{14} (1)$$

$$= 0.94 - 0.4688 - 0.429 = 0.042$$

Wind

weak	strong
{D ₁ , D ₃ , D ₄ , D ₅ , D ₆ , D ₉ , D ₁₀ , D ₁₃ }	{D ₂ , D ₇ , D ₁₁ , D ₁₂ , D ₁₄ }
[6 Yes, 2 No]	[3 Yes, 3 No]
Yes	No

Naive Bayes

- text classification
- It has high dimensional dataset.
- It is based on bayes theorem.
- ~~It is a probabilistic classifier.~~
- It is a probabilistic classifier.
- It acts to predict on the basis of probability of an object.

$$P(A|B) = \frac{P(B|A)P(A)}{P(B)}$$

where P(A|B), - Posterior probability.

P(B|A) = Likelihood probability.

P(A) = prior probability.

P(B) = Marginal probability.

Working:

e.g: weather condition dataset.

Day	Outlook	Temp	Humi	Wind	Play game
D ₁	S	H	Hi	W	N
D ₂	S	H	Hi	S	N
D ₃	OC	H	Hi	W	Y
D ₄	R	H	Hi	W	Y
D ₅	R	C	N	W	X
D ₆	R	C	N	S	N
D ₇	OC	C	N	S	Y
D ₈	S	M	Hi	W	N
D ₉	S	C	N	W	Y
D ₁₀	R	M	N	W	Y
D ₁₁	S	M	N	S	X
D ₁₂	OC	M	Hi	S	X
D ₁₃	OC	H	N	W	Y
D ₁₄	R	M	Hi	S	N

target variable is 'play'
using the dataset we need to
decide a player can play or not.

Steps:

- 1) Convert the given dataset into a frequency table.
- 2) Generate the likelihood table by finding the probability.
- 3) Use Bayes's theorem to calculate posterior probability.

PS

If the weather is summer then should the player play game or not.

sol Consider the dataset.

Outlook	Rain	Summer	Overcast
Play	yes	yes	yes

Freq:

weather	yes	No
Summer	2	3
Rain	3	2
Overcast	4	0
Total	9	5

likelihood:

Weather	Yes	No
Summer	2	3
Rain	3	2
Overcast	4	0
	9/14	5/14

$= 0.642 = 0.357$

Apply Bayes's theorem.

$$\begin{aligned}
 P^A_B(\text{yes} | \text{summer}) &= \frac{P(\text{summer} | \text{yes}) P(\text{yes})}{P(\text{summer})} \\
 &= \frac{\left(\frac{2}{9}\right) \left(\frac{9}{14}\right)}{\left(\frac{2}{14}\right) \left(\frac{9}{14}\right)} \\
 &= \frac{(0.22)(0.357)}{0.142} \\
 &= \frac{0.07854}{0.142} \\
 &= \frac{\left(\frac{2}{9}\right) \left(\frac{9}{14}\right)}{\left(\frac{5}{14}\right)} \\
 &= \frac{(0.22)(0.142)}{0.357} = 0.085
 \end{aligned}$$

Linear regression:

It is a statistical technique that predicts a continuous value based on one (or) more independent attributes

e.g: to predict the housing price based on living data.

- 1) How to ~~Data~~ A complement applicant in Data mining.
 2) How to Data mining in Data science.

The linear regression aims to predict the continuous value, it cannot be applied directly to the classification task.

Evaluation: Suppose there are n tuples which are represented as p attributes

as $x_i = \{x_{i1}, x_{i2}, x_{i3}, \dots, x_{in}\}$ continuous variable y_i where $i=1$ to n . To learn the mapping of x_{in} variables to y_i variables $\hat{y}_i = \omega^T x_i + b$ is

called as

$$\hat{y}_i = \sum_{j=1}^n \omega_j x_{ij} + b$$

where $\hat{y}_i$ is the predicted output for i th tuple here $\omega = (\omega_1, \dots, \omega_n)^T$ is the n dimensional weight vector.

b is the bias scalar. One of the common linear regression model is "least square regression", follows loss function as

$$L(\omega, b) = \sum_{i=1}^n (y_i - \hat{y}_i)^2$$

$$= \sum_{i=1}^n (y_i - [\omega^T x_i + b])^2$$

so the best weight vector (ω) and bias scalar (b) will minimize the loss function $L(\omega, b)$. It measures the sum of squared difference between the predicted output value $\hat{y}_i$ and actual observed value y_i . so the optimal weight

(ω) will be given as

$$\omega = \frac{\sum_{i=1}^n (y_i - \hat{y}_i)^2}{\sum_{i=1}^n x_i^2 - \frac{1}{n} \left(\sum_{i=1}^n x_i \right)^2}$$

and the bias scalar function,

$$b = \frac{1}{n} \sum_{i=1}^n (y_i - \omega x_i)$$

where the average observed value

$$\bar{y} = \frac{1}{n} \sum_{i=1}^n y_i$$

among n training tuples

eg

Index(i)	1	2	3	4
Attributes(x_i)	1	3	5	7
Output(y_i)	4	10	14	16

(a) training tuples.

find the least square regression model

$$y = \omega x + b$$

Sol:

$$\hat{y}_i = \frac{\sum_{i=1}^n y_i}{n}$$

$$= (y_1 + y_2 + y_3 + y_4) / 4 = (4 + 10 + 14 + 16) / 4$$

$$= 44 / 4 = 11$$

$$\therefore \sum_{i=1}^n x_i (y_i - \hat{y}_i) = 1(4-11) + 3(10-11) + 5(14-11) + 7(16-11)$$

$$= 40$$

$$\therefore \frac{1}{n} \sum_{i=1}^n (x_i)^2 = \frac{1}{4} \left(\sum_{i=1}^n (1+3+5+7)^2 \right)$$

$$= \frac{(16)^2}{4} = \frac{256}{4} = 64$$

Therefore the optimal weight

$$\omega = \frac{\sum_{i=1}^n x_i (y_i - \hat{y}_i)}{\sum_{i=1}^n x_i^2 - \frac{1}{n} \left(\sum_{i=1}^n x_i \right)^2}$$

$$w = 40$$

$$\sum_{i=1}^n x_i^2 = 1^2 + 3^2 + 5^2 + 7^2 = 1 + 9 + 25 + 49 = 84$$

$$w = \frac{40}{84 - 64} = \frac{40}{20}$$

$$w = 2$$

Optimal bias scalar

$$b = \frac{\sum_{i=1}^n (y_i - wx_i)}{n}$$

$$b = \frac{12 - 2(1)}{5} + \frac{10 - 2(3) + 14 - 2(5)}{5} = \frac{16 - 2(1)}{5} = \frac{2 + 4 + 4 + 2}{5} = 12$$

$$b = \frac{12}{4} = 3$$

So the final answer is $y_i = wx_i + b$

$$y_i = 2x_i + 3$$

Index (i)	1	2	3	4	5
Attributes (x_i)	1	2	3	4	5
Output (y_i)	12	18	26	32	38

find the least square regression.

Sol

$$\hat{y} = \frac{\sum_{i=1}^n y_i}{n}$$

$$= (y_1 + y_2 + y_3 + y_4 + y_5) / n$$

$$= (12 + 18 + 26 + 32 + 38) / 5 = \frac{126}{5} = 25.2$$

$$\sum_{i=1}^n x_i (y_i - \hat{y}) = 1(12 - 25.2) + 2(18 - 25.2) + 3(26 - 25.2) + 4(32 - 25.2) + 5(38 - 25.2)$$

$$= 66$$

$$\therefore \frac{1}{n} \sum_{i=1}^n (x_i)^2 = \frac{1}{5} (1^2 + 2^2 + 3^2 + 4^2 + 5^2) = \frac{1}{5} (1 + 4 + 9 + 16 + 25) = \frac{55}{5} = 11$$

Therefore the optimal weight.

$$w = \frac{\sum_{i=1}^n x_i (y_i - \hat{y}_i)}{\sum_{i=1}^n x_i^2 - \frac{1}{n} \left(\sum_{i=1}^n x_i \right)^2}$$

$$\sum_{i=1}^n x_i^2 = 1^2 + 2^2 + 3^2 + 4^2 + 5^2 = 1 + 4 + 9 + 16 + 25 = 55$$

Sol

$$\hat{y} = \frac{\sum_{i=1}^n y_i}{n}$$

$$= (y_1 + y_2 + y_3 + y_4 + y_5) / n$$

$$= (12 + 18 + 26 + 32 + 38) / 5$$

$$= 126 / 5 = 25.2$$

$$\begin{aligned} \therefore \sum_{i=1}^n x_i (y_i - \hat{y}) &= 1(12 - 25.2) + 2(18 - 25.2) + 3(26 - 25.2) + 4(32 - 25.2) + 5(38 - 25.2) \\ &= -1.32 + -1.44 + 0.24 + 2.72 + 6.4 \\ &= 6.6 \end{aligned}$$

$$\therefore \frac{1}{n} \sum_{i=1}^n (x_i)^2 = \frac{1}{5} (1^2 + 2^2 + 3^2 + 4^2 + 5^2) = \frac{1}{5} (15) = 3$$

Therefore the optimal weight.

$$w = \frac{\sum_{i=1}^n x_i (y_i - \hat{y}_i)}{\sum_{i=1}^n x_i^2 - \frac{1}{n} \left(\sum_{i=1}^n x_i \right)^2}$$

$$\sum_{i=1}^n x_i^2 = 1^2 + 2^2 + 3^2 + 4^2 + 5^2 = 55$$

$$w = \frac{6.6}{\frac{45-55}{55-45}} = \frac{6.6}{10} = 0.66$$

$$w = 0.66$$

Optimal bias scalar,

$$b = \frac{\sum_{i=1}^n (y_i - wx_i)}{n}$$

$$b = \frac{1.2 - (0.66)(1) + 1.8 - (0.66)(2) + (2.6) - (0.66)(3) + 3.2 - (0.66)(4) + 3.8 - (0.66)(5)}{5}$$

$$b = \frac{2.7}{5} = 0.54$$

Logistic regression:

→ Binary classification

→ Sigmoid function.

It takes input value as independent variable and produces the probability value will be 0 (or) 1.

e.g

class 0 and class 1

→ Threshold value ≤ 0.5

→ If the value of logistic function for an input is greater than 0.5 then it belongs to class 1.

Rules:

1) A logistic regression predicts the output of a category of dependence variables. So the outcome be categorical or discrete value.

2) It can be either 0 (or) 1.

True (or) False (or) Yes (or) No.

3) It gives probability value ~~to~~ between 0 and 1

A) 'S' shape logistic function. This logistic function is known as Sigmoid function.

The results are calculated based on likelihood rather than absolute values.

Types of logistic regression:

1) Binary logistic regression:

It has two possible dependent variables.

e.g 1) pass / fail.

2) Yes / No.

3) 1 / 0.

2) Multi-nominal logistic regression:

In this, multiple classes can be classified as

* Classifying text into language.

* Predicting whether the student is going to college or not.

3) Ordinal logistic regression.